Pretrain the barrel across regions

Today each region's NN1/NN2 pair is trained from scratch on its own events. The proposal is to pretrain on the endcap's much larger sample and then fine-tune on the barrel.

NOW · PER REGION

```
barrel ← barrel events
endcap ← endcap events
```

- \+ No domain shift: each net sees one detector region.
- \+ Simple, and it matches published ATLAS practice.
- − The barrel learns from about 2,280 background events only.

PROPOSED · ACROSS REGIONS

```
pretrain ← endcap events
fine-tune ← barrel events
```

- \+ The barrel net starts from features learned on about $11\times$ more background events.
- \+ Same signal model in both regions (HSS, same mass points).
- − Each region sees a different slice of that signal through different reconstruction, so the signal/background rule may not carry over (tested by T1, T3).
- − The endcap's background dependence could bias barrel ABCD closure.

BOTH · SHARED TRUNK

```
one trunk ← both regions
stems + heads ← per region
```

- \+ What both regions share is learned from all the events.
- \+ What the detector does differently stays region-specific.
- − It has the same bias risk as across-region and needs a region token.

The test of either route is the same: does the barrel get a better classifier and better ABCD closure than training from scratch, **without** importing the endcap's biases?

Published region split: ATLAS, arXiv:2503.20445. Barrel/endcap reconstruction: ATLAS, arXiv:1311.7070. Counts: BASELINE files.

What pretraining and fine-tuning look like

NN1 and NN2 each get this structure and keep their own feature sets. They still train as a pair, as today, so the DisCo term between them still applies.

① PRETRAIN · ON THE ENDCAP



- **Data:** all endcap events (no barrel event ever enters).
- **Loss:** BCE, or BCE + α DisCo. No closure terms, which would encode the endcap's ABCD plane.
- **Trains:** everything.
- **Keeps:** the trunk weights. The stem and head are thrown away.

② FINE-TUNE · ON THE BARREL



- **Data:** barrel training folds only, using the same $k = 5$ folds and paired seeds as from-scratch training.
- **Loss:** the locked barrel loss, with whatever DisCo and closure terms BASELINE locks.
- **Trains:** the new stem and head at the normal learning rate, and the trunk at a lower one (or frozen as a variant).
- **Out:** barrel NN1/NN2 scores feed the ABCD plane as today.

trained in pretraining · carried over and adjusted · new for the barrel

Why this order: the trunk learns what separates LLP decays from background on the endcap's ~25k background events. The barrel's few events then only have to fit the barrel-specific stem and head and **nudge** the trunk, instead of learning everything from scratch.

Per-feature scaling is fitted separately in each region. The per-region variant has the same shape but pretrains on barrel events, e.g. by predicting masked features.

Pretraining vs. fine-tuning: what differs

Same network and same training loop. What differs is where training starts, what data it sees, what may move and how far, and what the result is for.

	PRETRAINING	FINE-TUNING
STARTS FROM	random weights	the pretrained weights
DATA	big sample: endcap, ≈24,850 bkg	small sample: barrel training folds
WHAT TRAINS	everything	new barrel stem + head fully; trunk gently, or frozen
LEARNING RATE	normal	low on the trunk, normal on new layers
LOSS	BCE, or BCE + α DisCo	full locked barrel loss, closure terms included
LENGTH	long: many events, many epochs	short: early stopping comes quickly
RESULT	intermediate: trunk kept, rest discarded	the deliverable: barrel NN1/NN2 → ABCD

IN SYMBOLS

Pretraining: $\theta_{\text{pre}} = \arg\min_\theta \mathcal{L}_{\text{pre}}(\theta;\, D_E), \quad \theta_0 \sim \text{random}$

Fine-tuning: $\theta^\star = \arg\min_\theta \mathcal{L}_{\text{ft}}(\theta;\, D_B), \quad \theta_0 = \theta_{\text{pre}}$

With few barrel events and a low learning rate, $\theta^\star$ stays near θ_{pre}. This acts roughly like adding $\lambda\|\theta - \theta_{\text{pre}}\|^2$ to the barrel loss, a penalty used explicitly as L2-SP.

The point: the barrel data only *adjusts* what the endcap taught. **The risk:** whatever the endcap taught, including its background dependence, persists unless the barrel data pushes it out. That's why closure terms act at fine-tune and why the bias-import check exists.

THE DIAL: HOW FAR FINE-TUNING MAY MOVE

frozen trunk — no movement → **low LR** — a little → **full LR** — ≈ scratch, better init

Discriminative fine-tuning: Howard & Ruder, arXiv:1801.06146. L2-SP: Li, Grandvalet, Davoine, arXiv:1802.01483. Counts: BASELINE files.

Why: the barrel is data-starved

BACKGROUND EVENTS

≈2,280

barrel

≈24,850

endcap · about 11× more

WHAT A SMALL SAMPLE COSTS

- The networks fit noise, and results move from seed to seed.
- The ABCD closure test loses power: it can pass because its error bars are too wide to fail.

WHAT PRETRAINING OFFERS

- **A better starting point.** Useful features are already in place, so the small sample spends its information on the task instead of on the basics.
- **Steadier training.** There are fewer directions in which to fit noise, so we expect less spread across seeds.
- **Precedent in HEP.** Fine-tuning a pretrained event classifier helps most when the target sample is small, and transfer between domains cuts the target data needed.

The gain is limited by what the pretraining sample knows that the barrel sample doesn't.

Counts: BASELINE files. Ho, Roberts, Han, Wang, arXiv:2412.10665 · Schott, Flek, arXiv:2605.07471 (both on the STAVE page).

Pretrain per region, or across regions?

PER REGION

`barrel → barrel  ·  endcap → endcap`

Pretrain on the region's own events (e.g. self-supervised on background: predict masked features), then fine-tune signal vs background in the same region.

- \+ No domain shift: same detector, same features, nothing to map.
- \+ No risk of importing another region's background dependence.
- \+ Acts as a regularizer: more training targets per event and steadier gradients.
- − **No new events.** The effective sample size is unchanged, so it can stabilize the barrel but can't add statistics.

ACROSS REGIONS

`endcap → barrel  ·  or one trunk for both`

Pretrain on the endcap (or on both together), then fine-tune on the barrel.

- \+ **About $11\times$ more background events.** This is the only route that adds real statistics to the barrel.
- \+ Same signal model in both regions (HSS, same mass points).
- − Each region sees a different slice of that signal through different reconstruction: barrel chambers sit in the toroid field, endcap chambers don't, and the vertex algorithms differ. The signal/background rule has to be tested (T1, T3), not assumed.
- − Only features with a counterpart in the other region can share weights.
- − The endcap's background dependence could carry into the barrel and **bias ABCD closure**.

Per-region pretraining is safe but can only **stabilize**. Across-region pretraining can **add statistics**, but it has to be shown not to import bias.

"Masking adds targets per event, not events": LEXICON goal set mk2 §6, §11.

What it takes: a guesstimate

STEP	WHAT GETS BUILT / RUN	BUILD	COMPUTE	CAN START
SETUP	Private STAVE repo; transfer fields in the BASELINE run spec (`init`, `pair_seed`, …)	1 d	—	NOW
STAGE 0 Feature map	Role table (barrel ↔ endcap) and shared / shifted / region-only classes on BASELINE files	1–2 d	minutes	NOW · KJ SIGN-OFF
STAGE 1 Screens	T1 zero-shot, T2 domain classifier, T3 sign check (plain BCE nets); T4 copula reuses BASELINE cells	3–4 d	~200 small runs · hours	NOW · T4 after Stage A
STAGE 2 Build	Stem → trunk → head model; pretrain → swap stems → fine-tune; frozen-trunk and shared-trunk variants; W (masked-feature pretraining); leakage tests	5–7 d	—	NOW
STAGE 2 Run	9 cells × 10 seeds × 5 folds ≈ 450 fine-tunes, plus ≈ 70 pretraining runs	1 d	≈ 1–2 GPU-days	AFTER BASELINE LOCK
STAGE 3 Grade	Paired Δ vs S0 and A4, bias import, 20-seed confirmation of the best arm	2–3 d	≈ 0.5 GPU-day	with Stage 2
WRITE-UP	Results deck and public STAVE page update	1–2 d	—	end

≈ 14–20 working days of build and analysis, plus **≈ 2 GPU-days** of unattended runs. That's about **3–4 focused weeks**, or 5–6 calendar weeks alongside other projects.

Critical path: the BASELINE lock (S0 is the locked barrel cell) and KJ's sign-off on the role table. Everything up to the Stage 2 build can start now.

Guesstimate. Compute assumes ~1–3 min per barrel run and ~5–10 min per endcap pretraining run on the RTX 4070 SUPER with early stopping; check against BASELINE run logs. Cells: S0 scratch · A4 random init · W within-region · A1 fine-tune · A3 frozen trunk · A2 shared trunk, with A1–A3 at two pretraining losses.

How we'd know it worked

STAGE 1 · SCREENS BEFORE ANY FINE-TUNING

TEST	PROCEDURE	FAIL MEANS
T1 Zero-shot	Train on the endcap using $\mathcal{S}_k$ only, evaluate on barrel out-of-fold, and compare with a barrel-trained net on the same roles (paired seeds)	AUC near 0.5: concept shift, so don't transfer that net
T2 Domain clf.	Classify barrel vs endcap on scaled $\mathcal{S}_k$, for background and for each signal mass point	High AUC: the joint is shifted, so drop the top features from $\mathcal{S}_k$
T3 Sign check	Single-feature signal-vs-background AUC in each region	The two regions fall on opposite sides of 0.5: drop the feature from the trunk
T4 Copula	Train the F1 / F2 cells per region; compare the Legendre spectra of background NN1–NN2 scores on common orders $j, k \leq 4$	Spectra differ: the fine-tune needs closure pressure, and bias import decides

Thresholds are pre-registered before running. **Go to Stage 2** only if, for each net, T1 passes and $\mathcal{S}_k$ still has enough features to matter after the T2/T3 removals.

STAGE 3 · PAIRED GRADING vs S0

Report AUC (NN1, NN2, worst mass point), $\%|nc|$ per band, frac2sig, pull and the Legendre totals, each as a seed-by-seed Δ against S0. At ≈456 events per fold, **only paired differences carry evidence**.

Bias import (new): on the common Legendre coefficients,

$$d_{\text{ft}} = \|\theta_{\text{arm}} - \theta_E\| \quad \text{vs} \quad d_0 = \|\theta_{S0} - \theta_E\|.$$

If $d_{\text{ft}} < d_0$ beyond the seed spread, the barrel's background dependence has moved toward the endcap's.

Verdict. Transfer is useful if an arm beats *both* S0 and A4 (paired) with no bias-import flag. Beating S0 but not A4 is an architecture effect, not transfer.

NOW · NEXT · OPEN

NOW Stage 0 classes + role table draft, T1–T3 · AFTER BASELINE STAGE A T4 · AFTER LOCK Stage 2 arms

Open: D1 adopt the plan · D2 T1 threshold · D3 P-BCE and/or P-α · D4 A5 · D5 role-table sign-off · D6 bias import in the BASELINE grader · endcap counterparts for barrel-only NN1 inputs

Source: STAVE_endcap_to_barrel_pretraining_design_mk1 §5–9. LEXICON P-b (token-level pretraining) reuses the Stage 0 role table.

Sources

LITERATURE

1. J. Ho, B. R. Roberts, S. Han, H. Wang (2024). Pretrained Event Classification Model for High Energy Physics Analysis. arXiv:2412.10665 slide 4
2. M. Schott, L. Flek (2026). Transfer Learning Across Fast- and Full-Simulation Domains in High-Energy Physics. arXiv:2605.07471 slide 4
3. ATLAS Collaboration (2025). Search for events with one displaced vertex from long-lived neutral particles decaying into hadronic jets in the ATLAS muon spectrometer. arXiv:2503.20445 slide 1
4. ATLAS Collaboration (2014). Standalone vertex finding in the ATLAS muon spectrometer. JINST 9 P02001. arXiv:1311.7070 slides 1, 5
5. G. Kasieczka, B. Nachman, M. D. Schwartz, D. Shih (2021). ABCDisCo: Automating the ABCD Method with Machine Learning. Phys. Rev. D 103, 035021. arXiv:2007.14400 slides 2, 3
6. G. Kasieczka, D. Shih (2020). DisCo Fever: Robust Networks Through Distance Correlation. Phys. Rev. Lett. 125, 122001. arXiv:2001.05310 slides 2, 3
7. J. Howard, S. Ruder (2018). Universal Language Model Fine-tuning for Text Classification. ACL 2018. arXiv:1801.06146 slides 2, 3
8. X. Li, Y. Grandvalet, F. Davoine (2018). Explicit Inductive Bias for Transfer Learning with Convolutional Networks. ICML 2018. arXiv:1802.01483 slide 3

PROJECT NOTES · 4 NOV 2026

- STAVE: endcap → barrel pretraining design, mk1 slides 2, 3, 6, 7
- BASELINE: data checks and transfer run spec, mk1 slides 6, 7
- LEXICON: goal set and model labels, mk2, §6 and §11 slide 5

DATA & PUBLIC PAGES

- Background event counts: BASELINE data files slides 1, 3, 4
- STAVE project page: ATLAS_Master_Project_Repo/#stave

Ho et al., Schott & Flek, the ATLAS searches and ABCDisCo are also listed on the STAVE project page.